MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1MDCECT100 Home Appliances and Troubleshooting

Duration: 1 hour **Maximum Marks: 35**

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C)*

Multiple Choice Questions Answer **35** Questions
Each question carries **one** mark

1. Which device is commonly used to measure voltage? K 1
   a)Ammeter b) Voltmeter c)Ohmmeter d)Wattmeter
2. According to Ohm's Law, which of the following is true? U 1
   a) $V = I \times R$ b)$I = V \times R$ c) $R = V \times I$ d) $V = I + R$
3. In a parallel circuit, how does the voltage across each component will measure? U 1
   a)It is different for each component b)It adds up to the total voltage
   c) It is zero for all components d)It is the same for each component
4. Which of the following describes a circuit with no break? K 1
   a) Open circuit b) Closed circuit c) Short circuit d) Series circuit
5. What happens to the current in a series circuit when more resistors are added? U 1
   a) Current increases b) Current decreases
   c) Current remains the same d) Current becomes zero
6. How will increasing the length of a wire affect its resistance? U 1
   a) Resistance decreases b) Resistance increases
   c) Resistance remains the same d) Resistance becomes zero
7. Power in an electrical circuit is defined as: K 1
   a) Voltage times resistance b) Current times resistance
   c) Voltage times current d) Voltage divided by current
8. In a circuit where voltage is constant, if the resistance is halved, what happens to the power? U 1
   a) Power doubles b) Power remains the same
   c) Power quadruples d) Power decreases
9. What type of load is typically connected to a single-phase system? K 1
   a) Large industrial motors b) Residential lighting and appliances
   c) High-power machines d)Transformers
10. What is the primary purpose of earthing in electrical systems? K 1
    a) To increase voltage b) To provide a path for fault current
    c) To improve conductivity d) To reduce energy consumption
11. What is the main symptom of a phase-to-phase fault? U 1
    a) Overheating of equipment b) Flickering lights
    c) Voltage imbalance between phases d) Complete power failure

12. Which color represents the earth wire in industrial wiring? K 1

a) Green/Yellow b) Red c) Black d) Brown

13. Which wire gauge would typically be used for lighting circuits? K 1

a) 10 AWG b) 12 AWG c) 14 AWG d)20 AWG

14. What type of cable accessory is used for securing cables? U 1

a) Cable ties b) Glands c) Connectors d) Switches

15. What is the primary purpose of cable management in electrical systems? K 1

a)Aesthetics b) Performance c) Organizing cables d) Reducing costs

16. Which of the following is NOT a benefit of good cable management? U 1

a) Enhanced airflow b) Reduced signal interference

c) Easier troubleshooting d)Slower data transmission

17. What does MCB stand for? K 1

a) Mini Control Box b) Maximum Circuit Breaker

c) Miniature Circuit Breaker d) Minor Circuit Box

18. How does an RCCB protect against electrical faults? U 1

a) By cutting off power when an imbalance is detected

b)By regulating voltage levels

c)By reducing the current flow d) By stabilizing the power supply

19. What is the main difference between an ELCB and an RCCB? U 1

a) ELCB is older and detects leakage through earth wire

b) ELCB works on magnetic principles

c) RCCB detects overloads d) ELCB can detect overload

20. Which tool is commonly used for insulation resistance testing during equipment inspection? K 1

a) Multimeter b) Megger c) Ammeter d) Voltmeter

21. What type of screwdriver has a flat, straight blade that fits into a straight screw slot? K 2

a) Phillips screwdriver b) Flathead screwdriver c) Hex screwdriver d) Torx screwdriver

22. What is the main function of opening pliers? U 2

a) To tighten screws and bolts b) To grip and twist objects with adjustable jaws

c) To cut wires d)To measure angles in construction

23. Which type of tweezer tip is best for handling small, delicate objects? K 2

a) Slanted tip b) Flat tip c) Fine pointed tip d) Round tip

24. Allen keys are commonly used in which of the following applications? K 2

a) Furniture assembly b)Bicycle maintenance c) Machinery assembly d) All of the above

25. Which part of a hammer is responsible for striking objects? K 2

a)The handle b)The head or face c)The claw d) The grip

26. What type of motion does a jigsaw blade perform? U 2

a)Rotating motion b)Back-and-forth (reciprocating) motion

c) Circular motion d) Vibrational motion

27. Which material is typically used to make punches due to its hardness and durability? K 2

a)Aluminum b)Plastic Stainless c) steel d)Copper

28. Which type of wrench is used to apply a turning force to pipe fittings? U 2

a)Allen wrench b)Torque wrench c) Pipe wrench d)Spanner wrench

29. Which of the following materials can a scriber be used to mark? U 2

a) Only wood b) Metal, wood, and plastic c) Only metal d) Only plastic

30. What type of voltmeter is commonly used for high voltage measurements? K 2

a) Analog b) Fluke c) Digital d) All of the above

31. When measuring current, how should an ammeter be connected? U 2

a) In series b) In parallel c) Across the load d) None of the above

32. What is the resistance should an ideal ammeter have? U 2

a) Infinity b) High c) Low d) Zero

33. What is the primary advantage of a clamp meter? K 2

a) Compact size b)Measures DC c) No need to disconnect wires d)All of the above

34. What does a very high resistance reading (e.g., "OL" or in the MΩ range) across the fuse terminals suggest? U 2

a) The fuse is in good condition b)The fuse has very low resistance

c)The multimeter is malfunctioning d) The fuse is blown

35. How do you check for continuity in a circuit? U 2

a) Use voltage mode b)Use resistance mode

c) Use current mode d) Use capacitance mode

36. If the multimeter reads 0 Ω when measuring a resistor, what does it indicate? U 2

a) The resistor is functioning properly b) The resistor is open

c)The multimeter is set to the wrong mode d) The resistor has a short circuit

37. How do you identify a failed potentiometer? K 2

a) No change in resistance b) Infinite resistance c) Steady voltage d) Short circuit

38. What happens to a capacitor when it is fully charged? K 2

a) High resistance b)Low resistance c) Infinite resistance d) Zero capacitance

39. How can you identify a shorted inductor? K 2

a) High resistance b) Low resistance c)High inductance d)Infinite resistance

40. How do you check a MOSFET for gate functionality? K 2

a) Measure voltage b)Use diode test mode c)Measure resistance d) All of the above

1 * 35 = 35